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About me

I am a first-year Ph.D. student in a joint training programme between **Beijing Normal-Hong Kong Baptist University (BNBU)** and **Great Bay University**. I received my M.Sc. in Biological Engineering from the **University of Saskatchewan** (2026), where I worked with Prof. Yen-Han Lin on machine learning approaches for redox potential-controlled very-high-gravity ethanol fermentation, and my B.Eng. in Biological Engineering from **Northwest A&F University** (2024).

My research sits where **computational biology** meets **machine learning**. My Ph.D. direction is **spatial multi-omics** and **AI for Science**: building deep learning methods that integrate spatial and single-cell measurements — and their paired imaging — into interpretable models of cells and tissues. I came to this from bioprocess engineering, where I built soft sensors and kinetic models that turned cheap online signals into quantitative, forward-looking estimates of metabolism; that work is what taught me to treat noisy biological data as something to be modelled, not merely measured.

Position: Ph.D. Student, Year 1 · Computer Science

Programme: Ph.D. in Computer Science · joint training

Research focus: Computational biology · spatial multi-omics · AI4Science

M.Sc.: University of Saskatchewan, 2026 (GPA 4.0/4.0)

B.Eng.: Northwest A&F University, 2024

I am open to research collaborations and discussions on computational biology, spatial and single-cell multi-omics, and AI for science.

Research 研究方向

My Ph.D. work moves into **computational biology**, with a focus on **spatial multi-omics** and **AI for Science**. I am interested in methods that integrate spatial and single-cell omics with imaging and other modalities, and in deep generative and graph-based models that turn such measurements into interpretable, testable biology — toward a quantitative, AI-driven view of cells and tissues.

Current focus 当前方向

Computational biology & bioinformatics; Spatial multi-omics & spatial transcriptomics; Single-cell omics;

Multimodal integration (omics × imaging); AI4Science & machine learning methods;
Graph neural networks & generative models; Interpretable, uncertainty-aware prediction

Prior experience 既往经历

Before moving into computational biology I worked on bioprocess engineering and kinetic modelling. That background is where my interest in data-driven methods started: fitting mechanistic models to sparse, noisy measurements, and quantifying what the data can and cannot support.

Fermentation process control; Kinetic modelling & parameter inference; Soft sensing & state estimation;
Bayesian uncertainty quantification; Bioenergy & cleaner biomanufacturing

Publications 学术成果

Bayesian inference on fermentation kinetics: Comparative analysis with frequentist approach

PUBLISHED

Xiang Li, Yen-Han Lin

Bioresource Technology, 2026, 444, 134012. IF 9.0, JCR Q1.

DOI → (<https://doi.org/10.1016/j.biortech.2026.134012>) ScienceDirect → (<https://www.sciencedirect.com/science/article/pii/S0960852426000933>)

Enhancing cleaner biomanufacturing of ethanol: A temporal deep learning soft sensor for ORP-driven metabolic forecasting in very-high-gravity fermentation

UNDER REVIEW

Xiang Li, Yen-Han Lin

Journal of Cleaner Production — under review, 2026.

Modified Hyperbolic Secant (MHS) Models for Fermentation Kinetics and Uncertainty

POSTER

Xiang Li, Yen-Han Lin

Poster, *YABEC 2026* (Asian Young Biotechnologists Network), University of Saskatchewan.

CV 简历

EDUCATION

Ph.D. Student, Computer Science 2026 – present

Beijing Normal-Hong Kong Baptist University (BNBU) & Great Bay University, Guangdong, China
Joint training programme. Year 1.

M.Sc., Biological Engineering

2025 – 2026

University of Saskatchewan, Saskatoon, Canada

GPA 4.0/4.0. Supervisor: Prof. Yen-Han Lin. Thesis: *Machine learning approaches for redox potential-controlled very high gravity ethanol fermentation.*

B.Eng., Biological Engineering

2020 – 2024

Northwest A&F University, Yangling, China

GPA 3.71/4.0, rank 3/63. Thesis: *Deep learning-based image processing of maize allele-specific chromatin interactions and regulation of imprinted genes.*

RESEARCH EXPERIENCE

Deep learning monitoring and forecasting system for ethanol fermentation

Jun. 2025 – present

- Fused sparse offline measurements with dense online ORP signals to build a high-resolution training set; developed a real-time state-estimation model (ORPSN, RNN) and a multi-horizon forecasting model (ORPSC, LSTM encoder-decoder) for metabolic dynamics.
- **Outcome:** Achieved high-accuracy soft sensing of ethanol production and glucose consumption rates; maintained accuracy over an 8-hour forecast horizon, providing a basis for proactive feed-forward control.

Bayesian inference for fermentation kinetics and uncertainty analysis

Jan. 2025 – Dec. 2025

- Built Bayesian hierarchical kinetic models for multiple fermentation systems with algorithmic and literature-informed priors; systematically compared Bayesian and frequentist inference for parameter identifiability and predictive stability under sparse data.
- **Outcome:** Resolved non-convergence of conventional methods in small-sample settings, giving robust kinetic parameter estimates with explicitly quantified uncertainty and improved biological interpretability.

Allele-specific chromatin interactions and imprinted gene regulation in maize endosperm

Mar. 2023 – May 2024

- Applied autoencoders and U-Net to denoise and enhance the resolution of Hi-C data from reciprocal crosses of maize inbred lines B73 and Mo17; constructed allele-specific chromatin interaction maps; partitioned A/B compartments and identified TADs genome-wide; analysed compartment distributions around imprinted genes.
- **Outcome:** Built an allele-resolved 3D chromatin architecture of maize endosperm and showed that the compartmental distribution of imprinted genes has no clear parent-of-origin dependence, indicating that chromatin reorganisation at the A/B compartment level is not the principal mechanism of imprinting regulation there.

Metagenomics and culturomics for tobacco biological control	Nov. 2021 – Nov. 2022
- Screened and validated antagonistic strains and assembled consortia; designed and ran field trials; performed 16S rRNA amplicon analysis of the tobacco rhizosphere microbiome. Outcome: Established a biocontrol strain library with good field efficacy and developed an effective composite microbial agent.	

RESEARCH VISITS & INDUSTRY

Visiting Student	Jan. 2026 – present
Shanghai Jiao Tong University, School of Life Sciences and Biotechnology	
Project: deep learning–based metabolic modelling of filamentous fungi and its industrial application.	
Sales Engineer	Sep. 2024 – Dec. 2024
Servicebio Technology Co., Ltd., Zhengzhou, China	
Technical support and product application for life–science research reagents.	

HONOURS & AWARDS

University of Saskatchewan CGPS 75th Recruitment Scholarship	Jan. 2025
National Scholarship for Encouragement (国家励志奖学金), Northwest A&F University	Nov. 2021
First Prize, National Undergraduate Life Sciences Competition (Innovation & Entrepreneurship)	Nov. 2021
Meritorious Winner (top 8%), Mathematical Contest in Modeling, COMAP, USA	Apr. 2021
First-Class Academic Scholarship, College of Life Sciences, Northwest A&F University	Dec. 2020

SKILLS

Python; PyTorch; TensorFlow / Keras; NumPy & pandas; Bayesian inference (MCMC); Kinetic modelling; Bioreactor operation & ORP control; HPLC analysis; R; Git